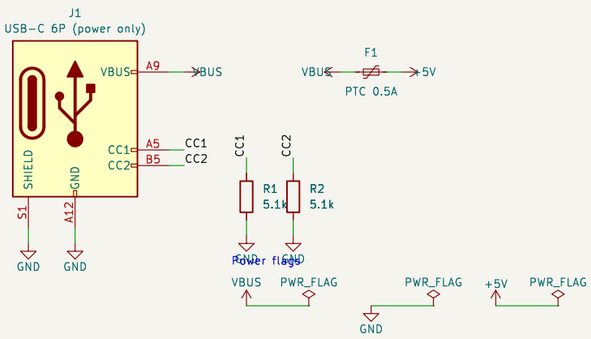
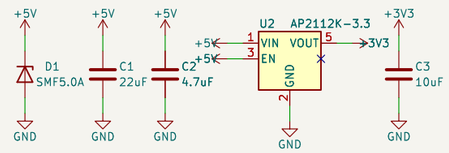
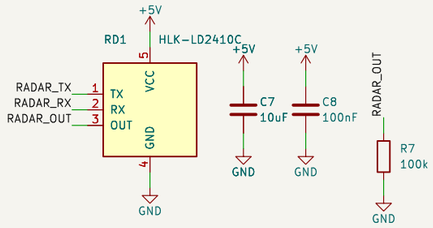
1. USB-C 5 V input (power only)



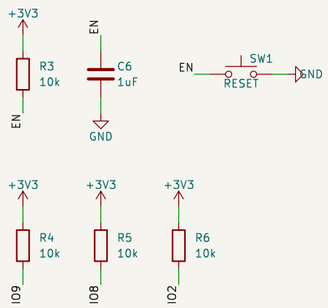
2. Protection + 3.3 V LDO



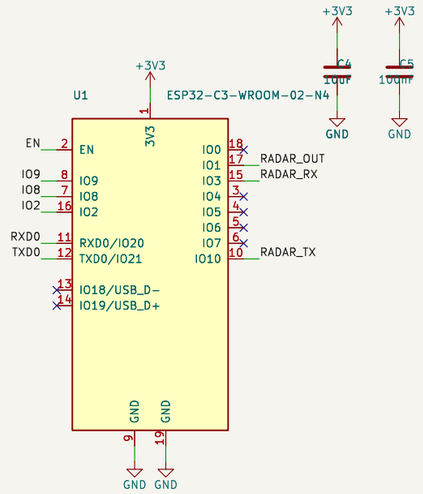
3. HLK-LD2410C radar (beside U1, antenna face up)



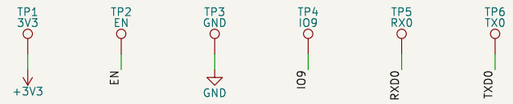
5. Reset / boot / strapping



4. ESP32-C3-WROOM-02-N4



6. Programming pads (pogo / solder, bottom side)



NOTES
1. Round PCB D45 mm, 2 layers, 1.6 mm. U1, RD1, C4, C5 on the top (room side); everything else on the bottom.
2. RD1 (HLK-LD2410C) sits flat below U1, not stacked: header insulator on the PCB, antenna face up, about 16 mm from the WiFi antenna zone. Solder the 5 pins from the bottom and trim them.
3. LD2410C pin order seen on the antenna face with the pins at the top: TX RX OUT GND VCC. Always check against the silkscreen of your module before soldering.
4. Radar UART1: RADAR_TX -> IO10 (ESP RX), IO3 (ESP TX) -> RADAR_RX, RADAR_OUT -> IO1. 256000 baud 8N1.
5. Flashing over UART0 pads RX0/TX0 (3.3 V USB-UART). Download mode: short IO9 to GND while tapping RST (or EN to GND). Power from USB-C or the 3V3 pad, not both.
6. USB-C is power only: CC1/CC2 each have their own 5.1k to GND. PTC + TVS on the 5 V input. AP2112K-3.3 (600 mA) for the ESP32-C3; the radar runs from 5 V directly.
7. Unused GPIO: IO0 IO4 IO5 IO6 IO7 IO18 IO19 (not routed on the PCB).
8. H1/H2: M2 mounting holes, PCB only (no symbol). No copper or metal within 15 mm of the antenna.

USB-C power only, UART0 pads for flashing, OTA afterwards
2-layer 1.6 mm, D45 mm. U1 + RD1 side by side on top, other parts bottom side

Sheet: /
File: c3_radar_r45.kicad_sch

Title: c3-radar r45: ESP32-C3 + HLK-LD2410C presence sensor, round 45

Size: A3	Date: 2026-10-19	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1